LAMBERT DE MONTE

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Email: lambert.demonte@kaust.edu.sa **Google Scholar:** Publications **GitHub:** Repositories

CURRENT POSITION

King Abdullah University of Science and Technology (KAUST) *Aug. 2026 – Present*
Remote Consultant and incoming Postdoctoral Research Fellow in Prof. Raphaël Huser's XSTAT group.

EDUCATION

PhD Statistics, University of Edinburgh, United Kingdom *Sep. 2022 – Jun. 2026*
Supervisors: Dr. Ioannis Papastathopoulos, Prof. Gabriele C. Hegerl
Thesis: *Geometric considerations for generative modelling of multivariate extremes*
Outcome: Successfully defended on June 29, 2026, with no corrections required.

Visiting PhD student, KAUST, Saudi Arabia *Feb. 2024 – May 2024 and Nov. 2025*
Supervisor: Prof. Raphaël Huser

MSc Mathematics and Statistics, McGill University, Canada *Sep. 2020 – June 2022*
Supervisor: Prof. Christian Genest
Thesis: *A spatial Gamma-Gamma model for the statistical postprocessing of ensemble weather forecasts*

BSc Mathematics and Computer Science, McGill University, Canada *Sep. 2017 – May 2020*

PROFESSIONAL EXPERIENCE

King Abdullah University of Science and Technology (KAUST) *Aug. 2026 – Present*
Remote consultant
- Developing machine learning methods for the spatial modelling of environmental extreme events.

Hydro-Québec Research Institute (IREQ) *Jan. 2022 – Aug. 2022*
Research intern and consultant – Mitacs-funded industry partnership (CAD $30,000)
- Developed spatial statistical methods for post-processing ensemble weather forecasts, targeting the extreme meteorological conditions that matter for energy production and logistics of operations.
- Worked with operational ensemble forecasts on Hydro-Québec's CASIR computing infrastructure.

DataHub, National Bank of Canada *May 2019 – Sept. 2020*
Data Science Intern
- Development and automation of the official DataHub credit card database.
- Statistical modeling of credit card use and behaviours.

PUBLICATIONS

Research
- De Monte, L. & Papastathopoulos, I. (2026), *A directional Poisson point process for univariate extremes*, Statistics & Probability Letters **235**, 110722.
- De Monte, L., Huser, R., Papastathopoulos, I., and Richards, J. (2025) *Generative modelling of multivariate geometric extremes using normalising flows*, arXiv preprint.
- Papastathopoulos, I., De Monte, L., Campbell, R. & Rue, H. (2023) *Statistical inference for radially-stable generalised pareto distributions and return level-sets in geometric extremes*, arXiv preprint.

Book chapter
- Auld, G., De Monte, L. & Papastathopoulos, I. (2026) Time Series in Extremes. In de Carvalho, M., Huser, R., Naveau, P., and Reich, B. J., editors, *Handbook on Statistics of Extremes*. Chapman & Hall/CRC, Boca Raton, FL.

Theses
- De Monte, L. *Geometric considerations for generative modelling of multivariate extremes.* PhD thesis. The University of Edinburgh, 2026.
- De Monte, L. *A Spatial Gamma-Gamma Model for the Statistical Postprocessing of Ensemble Weather Forecasts.* Master's thesis. McGill University, 2022.

AWARDS AND FUNDING

School of Mathematics, University of Edinburgh *Sep. 2023 – Jul. 2026*
- Full PhD Studentship: £27,600 tuition and £20,800 stipend per year.

Department of Mathematics and Statistics, McGill University *Sep. 2020 – Aug. 2022*
- MSc Studentship: $26,600 stipend for the duration of the masters.

Mitacs Canada *Jan. 2022*
- Funding for research on extreme value modeling in partnership with Hydro-Québec (CAD $30,000)

SOFTWARE (OPEN-SOURCE)

- R package geometricExtremes: creator and maintainer. Documented package (help pages, worked examples, citation file) for Bayesian inference in geometric extremes; maintained since 2023. Based on Papastathopoulos et al. (2023).
- Python code geometricExtremesNF: creator. Reference implementation of normalising-flow models for multivariate geometric extremes. Based on De Monte et al. (2025).
- R code TSExtremes: contributor. Reproduction code and data for the case study in Auld et al. (2026).

ACADEMIC SERVICE AND LEADERSHIP

Postgraduate representative, School of Mathematics (University of Edinburgh)
Representative of the postgraduate students of the *School of Mathematics* *Sept. 2024 – Oct. 2025*
- Intermediary between the students and the School administration.
- Organisation of social events within the School.
- Representing the *School of Mathematics* at the *Martingale PhD Navigator Programme* events for future scholars (December 2024 and 2025).

Scientific article reviewing
Environmetrics, Statistics and Computing

SCIENCE COMMUNICATION AND OUTREACH

- Richards, J. & De Monte, L. (2024) Review of "Risk Revealed: Cautionary Tales, Understanding and Communication" by P. Embrechts, M. Hofert & V. Chavez-Demoulin, JABES.
  Review of a book on communicating risk and uncertainty to non-specialist audiences.
- Genest, C. & De Monte, L. (2021) *À la recherche de la Rondurie*, Accromath, Volume 16, p.24–29.
  Article popularising mathematics for a French-language magazine read by students and teachers.

PRESENTATIONS AND POSTER SESSIONS

Invited
- Séminaires de sciences de la décision HEC Montréal, Montréal, Canada: *Multivariate extremal index and clustering in multivariate extremes: A case study of Dutch hydrological extremes.* *Oct. 2026*
- Generative AI Modelling for Extreme Events (GAME), Bologna, Italy: *Multivariate extremal index and clustering in multivariate extremes: A case study of Dutch hydrological extremes.* *Jun. 2026*
- CfS Annual Conference 2026, Edinburgh, United Kingdom: *Multivariate extremal index and clustering in multivariate extremes: A case study of Dutch hydrological extremes.* *Jun. 2026*
- Generative AI Modelling for Extreme Events, University of Edinburgh Generative AI Laboratory, Edinburgh, United Kingdom: *Generative modelling of multivariate geometric extremes using normalising flows.* *Jun. 2025*
- Statistics seminar, University of Exeter, Exeter, United Kingdom: *Geometric extreme value theory – A normalising flows approach*. *Feb. 2025*
- CMStatistics, London, United Kingdom: *Geometric extreme value theory – A normalising flows approach*. *Dec. 2024*
- CMStatistics, Berlin, Germany: *Bayesian inference for radially-stable distributions*. *Dec. 2023*
- BIRS-IMAG Modern Statistical and Machine Learning Approaches for High-Dimensional Compound Spatial Extremes, Granada, Spain (short presentation): *Geometric inference for Hüsler–Reiss random vectors*. May 2023
- Séminaires de sciences de la décision HEC Montréal, Montréal, Canada: *Multivariate extremes – A geometric Bayesian inference approach*. *Sep. 2023*

Contributed

- 14th International Conference on Extreme Value Analysis, University of North Carolina Chapel Hill, United States: *Generative modelling of multivariate geometric extremes using normalising flows* *Jun. 2025*
- 15th International Meeting on Statistical Climatology, Toulouse, France: *Flood risk modelling using geometric extreme value theory.* *Jun. 2024*
- CfS Annual Conference 2024, Edinburgh, United Kingdom (poster): *Multivariate radial Pareto distributions: a geometric approach to the statistical modelling of multivariate extremes.* *Jun. 2024*
- 13th International Conference on Extreme Value Analysis, Milan, Italy (poster): *Bayesian approach to geometric inference for multivariate extremes*. *Jun. 2023*

TEACHING AND MENTORING

Teaching Assistantships (University of Edinburgh)

Tutoring workshops. Marking of assignments and examinations. List of courses:

- MATH10093 – Statistical Computing *Jan. – May 2023, Jan. – May 2025, and Jan. – May 2026*
- MATH08065 – Computing and Numerics *Jan. – May 2026*
- MATH10096 – Applied Statistics *Jan. – May 2025 and Jan. – May 2026*
- MATH11199 – Python Programming *Sep. – Dec. 2025*
- MATH08066 – Probability *Sep. – Dec. 2023 and Sep. – Dec. 2025*
- MATH10064 – Multivariate Data Analysis *Jan. – May 2025*
- MATH11176 – Extended Statistical Programming *Sep. – Dec. 2023*
- MATH11187 – Generalised Regression Models *Sep. – Dec. 2023*
- MATH08051 – Statistics *Jan. – May 2023*

StatsBase Statistics Helpdesk (University of Edinburgh) *Nov. 2025*

Providing help to MSc students in Statistics for the Statistical Methodology and Theory of Statistical Inference courses.

Teaching Assistantships (McGill University)

Preparation of tutorial sessions, office hours for course-related questions. Marking of midterm and final exams, supervision of assignment marking. List of courses:

- MATH 324 – Statistics *Jan. 2022 – May 2022*
- MATH 323 – Probability *Sep. 2021 – Dec. 2021*
- MATH 203 – Principles of Statistics *Jan. 2021 – May 2021*

Statistics Helpdesk (McGill University) *Sep. 2020 – Dec. 2021*

Providing help to undergraduate students in Statistics and Probability. Theoretical and programming guidance for courses in the range MATH 2xx–5xx.

Undergraduate Project Supervision (McGill University)

Guidance of three students towards the completion of an undergraduate project in extreme value theory.

- MATH 470 – Honours Research Project *Jan. 2022 – May 2022*
- MATH 410 – Majors Project *Jan. 2021 – May 2021 and Sep. 2021 – Dec. 2021*

TECHNICAL SKILLS

Languages: R, Python, Julia, Bash, SQL, LaTeX.
Machine learning: PyTorch.
Weather and climate data: ensemble forecasts from ECMWF, CWAO (Environment and Climate Change Canada) and KWBC (NOAA/NCEP).
High-performance computing: Eddie (University of Edinburgh), Ibex (King Abdullah University of Science and Technology), CASIR (Hydro-Québec).
Cloud computing: Microsoft Azure (National Bank of Canada).

LANGUAGES

Bilingual: English, French.